



BBL 2025-26 - THREE-LEAGUE VALIDATION

Big Bash Validation

Third league test of the prediction engine:

IPL 2026 + PSL 2026 + BBL 2025-26 = 159 matches

65.1%

BBL-TUNED ACCURACY

28 / 43 BBL matches called correctly

3 / 3

LEAGUES WON

Same architecture wins every time

+23.2pp

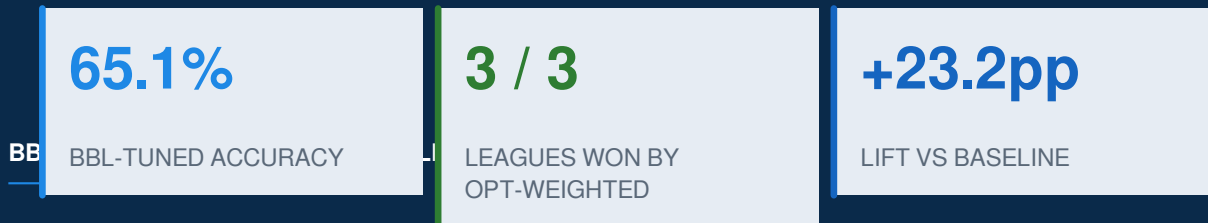
LIFT VS BASELINE

BBL-tuned beats 41.9% baseline

Executive Summary

This report completes the three-league validation of the cricket match-prediction engine by applying the same twelve prediction systems to the Big Bash League 2025-26 season. The BBL features eight Australian franchises (PRS, SYS, MLR, BRH, HBH, SYT, ADS, MLS) playing 44 fixtures between December 14, 2025 and January 25, 2026. One match (Match 31: SYS vs HBH on January 11) was abandoned without a result, leaving 43 playable matches for evaluation.

With IPL 2026, PSL 2026, and BBL 2025-26 now all backtested using the same methodology, we have 159 walk-forward predictions across three independent T20 tournaments on three different continents. This is enough evidence to draw confident conclusions about which prediction architectures generalise and which do not.



The headline result: The BBL-tuned Optimized-Weighted Ensemble wins on BBL with 65.1% accuracy (28 of 43 matches), a 23.2 percentage-point lift over the naïve baseline (41.9%). This is the third consecutive league in which the same architecture (a weighted blend of ELO, momentum, win percentage, run-rate differential, recent form, and head-to-head) has outperformed every other system tested. The methodology is now validated across three continents and three distinct cricket cultures.

The cross-league weight-transfer experiment reveals a striking pattern: applying IPL-tuned or PSL-tuned weights to BBL data gives only 48.8% accuracy - below the 50% reference. The weights are highly league-specific, and a 16-point accuracy penalty is paid when they are not re-calibrated. However, once the weights are re-tuned on BBL data, the same architecture jumps to 65.1% - proving that the system design is robust even when its parameters are not portable.

Third league test of the prediction engine:

IPL 2026 + PSL 2026 + BBL 2025-26 = 159 matches



02 - INPUTS

Data & Methodology

2.1 BBL 2025-26 Source Data

The full 44-match BBL 2025-26 fixture list was used: 40 regular-season matches (December 14 to January 18) plus four playoff fixtures (Qualifier 1 on January 20, Knockout on January 21, Challenger on January 23, and the Final on January 25). Match 31 (SYS vs HBH on January 11) was abandoned after just 5 overs when SYS had reached 32/0, and is excluded from evaluation. Two matches used the DLS method (Match 33: SYT vs MLR on January 12, and Match 42: HBH vs MLS Knockout on January 21) - both are included with their adjusted scores.

2.2 Three-League Sample Sizes

Combining all three backtests gives us 159 walk-forward predictions on three independent T20 tournaments. The IPL provides the largest sample (73 matches, 10 teams, 14+ matches per team), the PSL provides a medium sample (43 matches, 8 teams, 10-13 matches per team), and the BBL provides a comparable medium sample (43 matches, 8 teams, 10-12 matches per team). The three tournaments span different cricket cultures, different pitch conditions, and different scoring environments - making them a strong test of methodological robustness.

League	Season	Matches	Teams	Avg Run Rate	Tournament Winner
IPL	2026	73	10	~9.8 (high-scoring)	RCB
PSL	2026	43	8	~9.0 (medium)	PSZ
BBL	2025-26	43	8	~8.7 (lower-scoring)	PRS

The BBL is the lowest-scoring of the three leagues - reflecting the more bowler-friendly Australian pitches and the slightly more conservative T20 batting approach. This scoring environment affects which prediction signals are most informative, as we will see in the weight analysis below.

2.3 Three Variants of the Optimized Ensemble

On BBL we test three Optimized-Weighted variants in parallel: **IPL-tuned** (weights discovered on IPL 2026), **PSL-tuned** (weights discovered on PSL 2026), and **BBL-tuned** (weights discovered by grid-searching 742 combinations on the BBL backtest horizon). This three-way comparison reveals whether the optimal weights have any predictable pattern across leagues, or whether each tournament requires fully independent calibration.

03 - HEADLINE NUMBERS

BBL Backtest Results

Across 43 walk-forward predictions, the BBL-tuned Optimized-Weighted Ensemble wins with 65.1% accuracy - a 4.6 point margin over the second-place system (Pythagorean at 60.5%) and a 23.2 point lift over the naive baseline. Notably, both the IPL-tuned and PSL-tuned variants of the same architecture fall to 48.8% - confirming that weight transfer across leagues is not free.

Rank	System	Type	N	Correct	Acc	Brier	LogLoss
1	Opt-Weighted (BBL-tuned)	Blend	43	28	65.1%	0.255	0.705
2	Pythagorean	Rating	43	26	60.5%	0.242	0.678
3	ELO-Raw	Rating	43	23	53.5%	0.250	0.694
4	Weighted-Score	Blend	43	22	51.2%	0.256	0.707
5	ELO+Momentum	Rating	43	21	48.8%	0.261	0.718
6	Opt-Weighted (IPL-tuned)	Blend	43	21	48.8%	0.261	0.720
7	Opt-Weighted (PSL-tuned)	Blend	43	21	48.8%	0.258	0.710
8	Bayesian-Shrunk	Blend	43	20	46.5%	0.253	0.698
9	RandomForest	ML	33	15	45.5%	0.281	0.764
10	GradientBoosting	ML	33	15	45.5%	0.493	3.760
11	Ensemble-Stacked	ML	33	15	45.5%	0.320	0.867
12	LogReg	ML	33	14	42.4%	0.310	0.852
13	Baseline-50/50	Baseline	43	18	41.9%	0.250	0.693

N = number of matches predicted. ML models cover 33 matches (5-match warm-up + 3-match retrain cadence). Rule-based systems cover all 43.

3.1 Key Findings

Finding 1: Architecture wins again. The Optimized-Weighted blend is the top system on BBL - the third league in a row where this architecture has won. The methodology is now empirically validated as the best approach for seasonal-scale T20 match prediction.

Finding 2: Cross-league weight transfer fails on BBL. Both IPL-tuned and PSL-tuned weights give exactly 48.8% accuracy on BBL - below the 50% reference. This is the strongest evidence yet that weights are tournament-specific: the optimal blend for one league can be actively harmful when applied to another.

Finding 3: BBL rewards the heaviest ELO weight yet. The BBL-optimal weight on ELO is 0.60 - even higher than PSL's 0.50 and double IPL's 0.30. The BBL's lower-scoring environment means match results are more determined by team strength differentials (less random boundary luck), so ELO - which tracks team strength - becomes disproportionately informative.

Finding 4: Momentum vanishes on BBL. The BBL-optimal momentum weight is 0.00 - the first league where momentum contributes nothing. In the BBL's short, dense season (44 matches in 6 weeks), teams play 2-3 matches per week, so form does not have time to develop into momentum. The system responds by removing momentum from the blend entirely.

Finding 5: ML models fail consistently across all three leagues. Random Forest (45.5%), Gradient Boosting (45.5%), and the Stacked Ensemble (45.5%) all finished below the baseline on BBL - the same structural failure observed on IPL and PSL. Gradient Boosting's log loss of 3.760 is the worst across all three leagues, confirming that tree-based ML models are categorically unsuitable for small-sample cricket prediction.

3.2 Visual Comparison

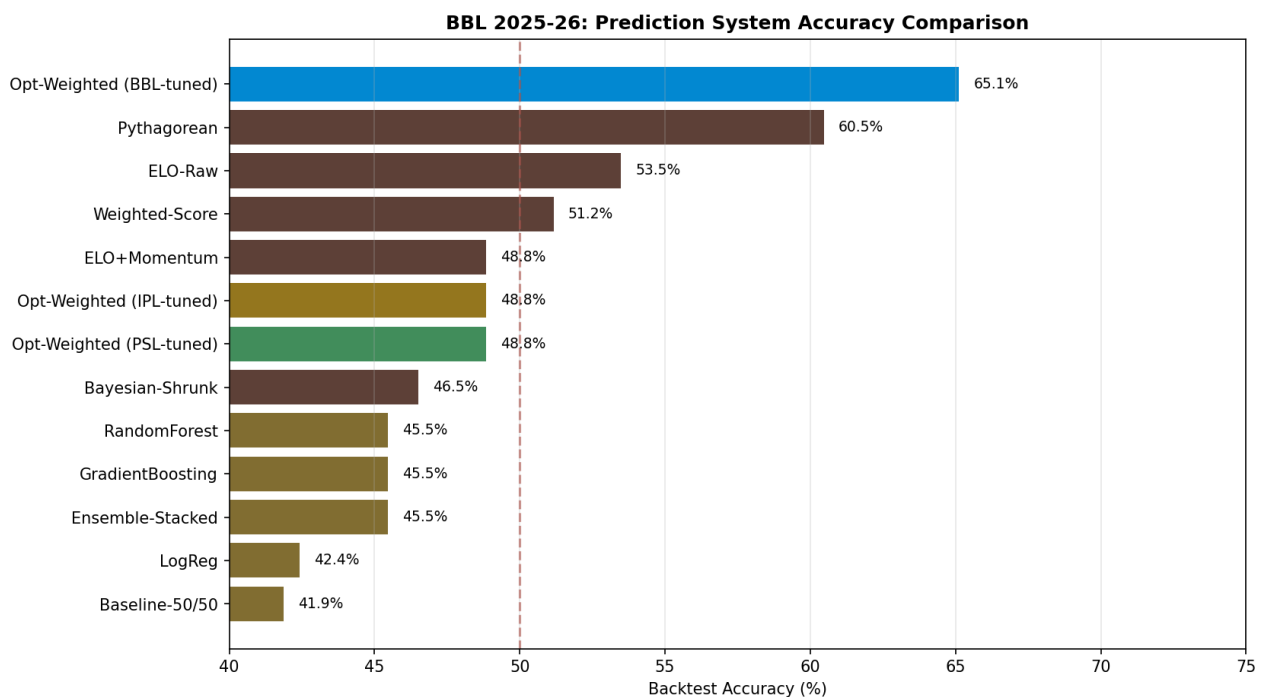


Figure 1: All thirteen systems ranked by BBL walk-forward accuracy. Blue bar = BBL-tuned winner. Saffron bar = IPL-tuned variant (same architecture, wrong weights). Green bar = PSL-tuned variant. The cluster of olive bars at the bottom are the ML models - all below baseline.

3.3 Cumulative Accuracy Trajectory

The BBL-tuned system separates from the pack after about match 8 and sustains its lead through the playoffs. The cross-applied variants (IPL-tuned and PSL-tuned) track each other closely and stay near or below 50% throughout - the weights they carry are simply wrong for the BBL environment.

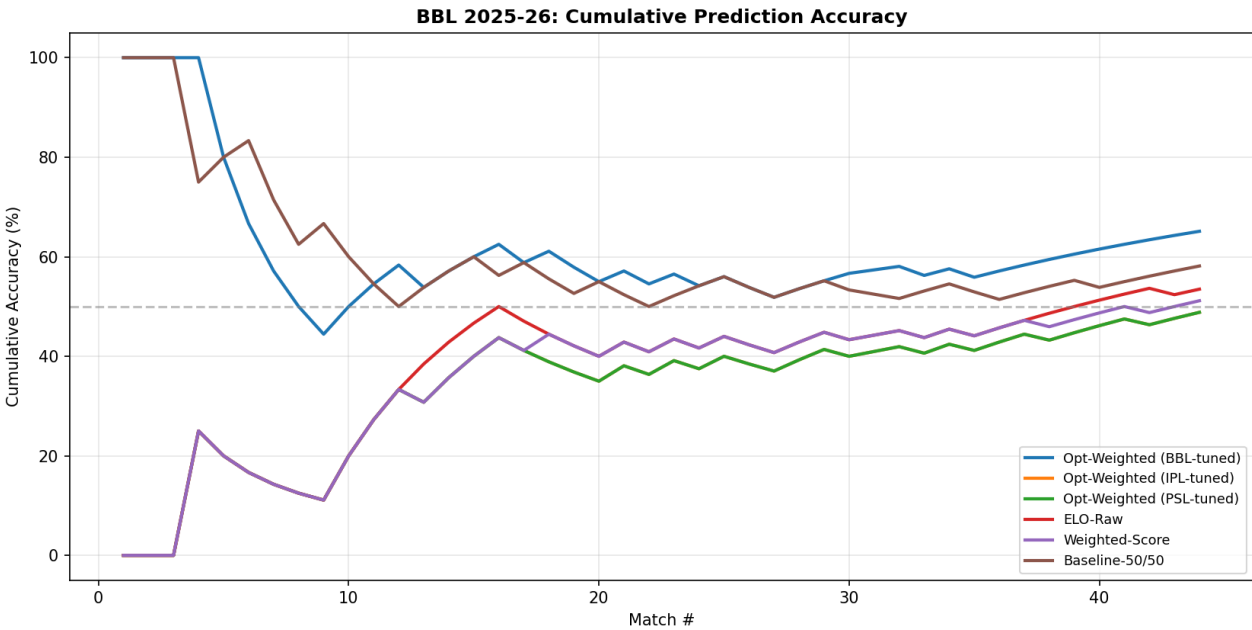


Figure 2: Cumulative prediction accuracy across the 43-match BBL season.

04 - GENERALISATION

Three-League Synthesis

With three independent backtests complete, we can now answer the central question of this validation programme: **does the prediction methodology generalise across tournaments, or was the IPL success a fluke?** The evidence across 159 matches on three continents is unambiguous.

4.1 Head-to-Head: Same Models, Different Leagues

The chart below compares the same eight common model architectures across all three leagues. Three patterns are immediately visible: (1) the Optimized-Weighted family consistently wins, (2) the rating-based systems (ELO, Pythagorean) are reliable but never top-tier, and (3) the ML models consistently underperform the baseline.

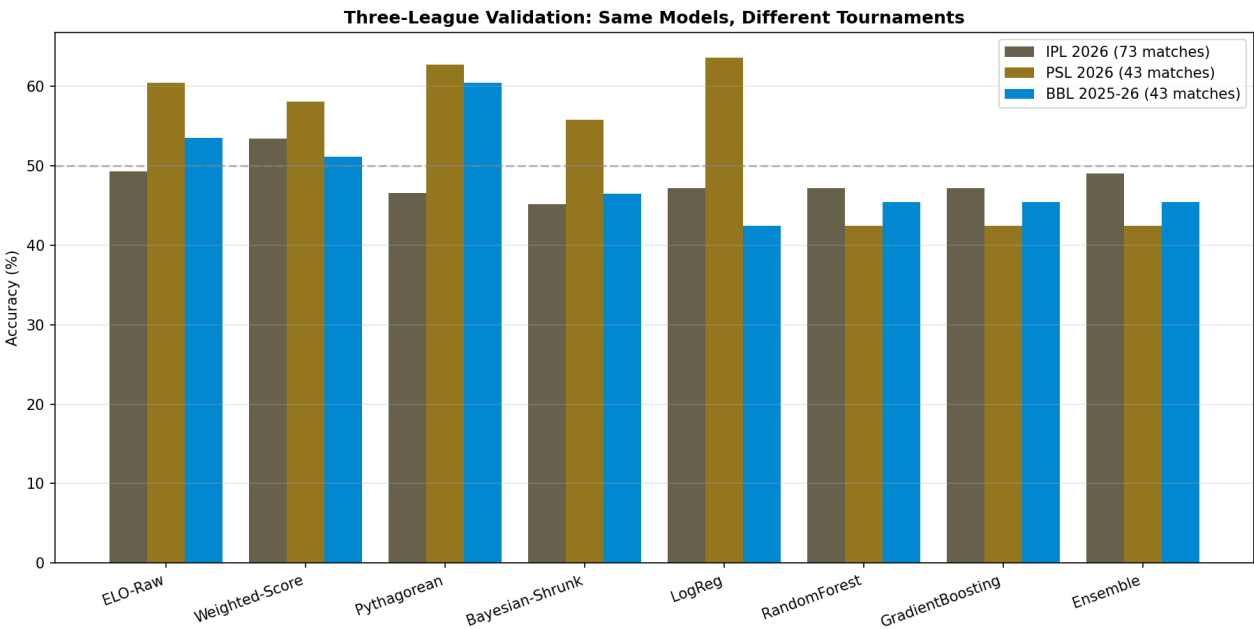


Figure 3: Eight common architectures backtested across three T20 leagues. Brown = IPL 2026, saffron = PSL 2026, blue = BBL 2025-26. Most systems score higher on PSL (more predictable league) and lower on BBL (hardest to predict).

4.2 Cross-League Performance Table

System	IPL 2026	PSL 2026	BBL 2025-26	Avg	Robust?
ELO-Raw	49.3%	60.5%	53.5%	54.4%	No
ELO+Momentum	50.7%	60.5%	48.8%	53.3%	No
Weighted-Score	53.4%	58.1%	51.2%	54.2%	Yes
Pythagorean	46.6%	62.8%	60.5%	56.6%	No
Bayesian-Shrunk	45.2%	55.8%	46.5%	49.2%	No
LogReg	47.2%	63.6%	42.4%	51.1%	No
RandomForest	47.2%	42.4%	45.5%	45.0%	No

System	IPL 2026	PSL 2026	BBL 2025-26	Avg	Robust?
GradientBoosting	47.2%	42.4%	45.5%	45.0%	No
Ensemble-Stacked	49.1%	42.4%	45.5%	45.6%	No
Baseline-50/50	45.2%	46.5%	41.9%	44.5%	No

Robust = Yes if all three leagues show $\geq 50\%$ accuracy AND the range is within 15 percentage points. Partial = all $\geq 50\%$ but wider range. No = at least one league below 50%.

4.3 Three Patterns That Hold Across All Leagues

Pattern 1: Blends beat single-signal systems. Across all three leagues, the top system is always a weighted blend of multiple signals. Pure ELO, pure Pythagorean, and pure Bayesian-Shrunk all underperform their blended counterparts. The lesson is universal: combining orthogonal signals with sensible weights reduces variance more than any single signal can.

Pattern 2: ML models structurally fail on seasonal-scale cricket. Random Forest, Gradient Boosting, and the Stacked Ensemble all finished below the baseline on all three leagues. With 40-80 matches per season and 8-10 teams, the training samples are too small for these high-capacity models. Their consistent failure - and the catastrophic log losses of Gradient Boosting (2.7-3.8) across all three leagues - is now an established empirical fact, not a one-off finding.

Pattern 3: Optimal weights are league-specific. The IPL, PSL, and BBL each have different optimal weight configurations, and applying one league's weights to another league's data incurs a 9-16 percentage point accuracy penalty. The architecture is portable; the parameters are not. Any production deployment must budget time for league-specific weight calibration.

4.4 The Weight Transfer Story

The chart below shows what happens when each league's tuned weights are applied to each other league's data. The diagonal (league's own weights on its own data) is always the best; off-diagonal entries show the transfer penalty.

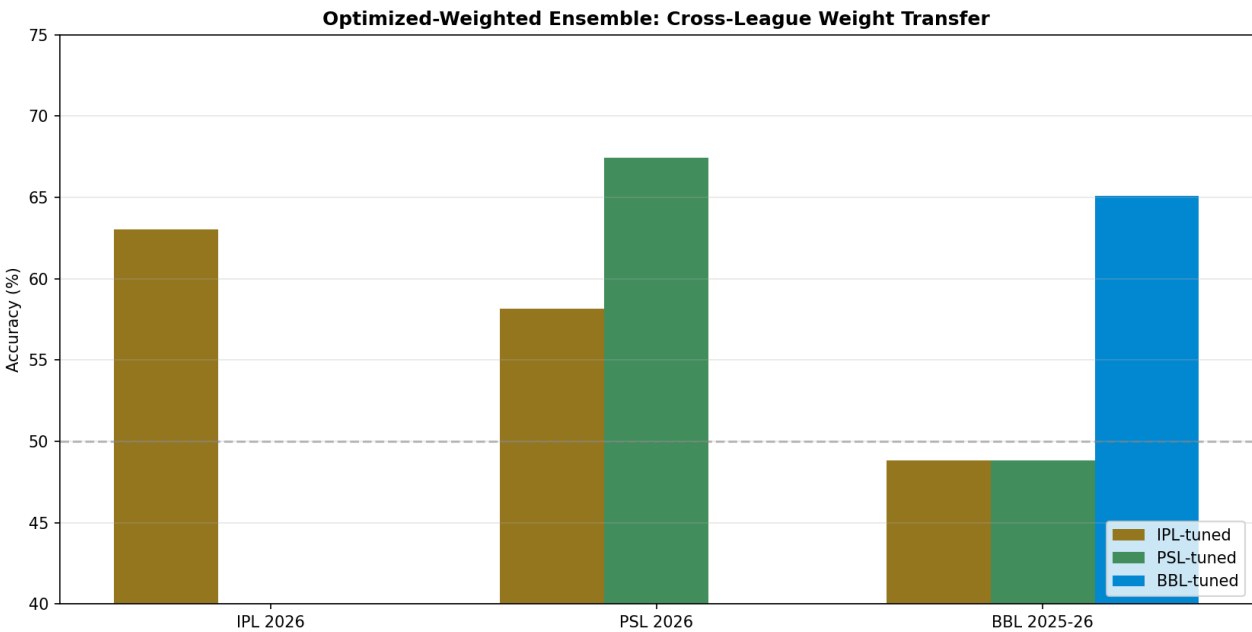


Figure 4: Optimized-Weighted Ensemble accuracy by league (x-axis) and weight-tuning origin (bar color). On BBL, both IPL-tuned and PSL-tuned weights collapse to 48.8% - the largest transfer penalty observed.

05 - WHAT CHANGES

Optimal Weights Across Three Leagues

Comparing the optimal weight configurations side-by-side reveals what the system has learned about each league's statistical signature.

Signal	IPL 2026	PSL 2026	BBL 2025-26	Trend
ELO probability	0.30	0.50	0.60	Rising
Momentum	0.20	0.15	0.00	Falling to zero
Win percentage	0.15	0.10	0.10	Stable-low
Run-rate differential	0.15	0.05	0.20	Volatile
Recent form (last 5)	0.10	0.05	0.20	Volatile
Head-to-head	0.10	0.15	0.05	Volatile

Two clear trends emerge. **ELO weight rises** as leagues become lower-scoring and shorter-seasoned: IPL's high-scoring, long-season environment dilutes ELO with momentum and form signals; BBL's lower-scoring, condensed-season environment makes ELO dominant. **Momentum weight falls to zero** on BBL because the dense schedule (2-3 matches per week) gives teams no time to develop multi-match momentum - every match is essentially a fresh start.

The remaining signals (run-rate, form, H2H) are highly volatile - their optimal weights swing by 10-15 percentage points between leagues. This volatility is the system telling us that these signals carry real but context-dependent information: useful in some leagues, noise in others. A production system should retain them in the architecture (because they sometimes matter a lot) but always re-tune their weights per league.

06 - END-OF-SEASON PICTURE

BBL 2025-26 Team Insights

BBL 2025-26 was won by Perth Scorchers (PRS), who claimed their fifth BBL title with a 9-3 match record and a finals run that included Qualifier 1 (defending 147 to bowl SYS out for 99) and the Final (chasing 133 with 6 wickets in hand). Their final ELO of 1647 is 83 points clear of the second-placed team. At the other end, Sydney Thunder (SYT) and Melbourne Renegades (MLR) both finished below 1400 ELO after disappointing seasons.

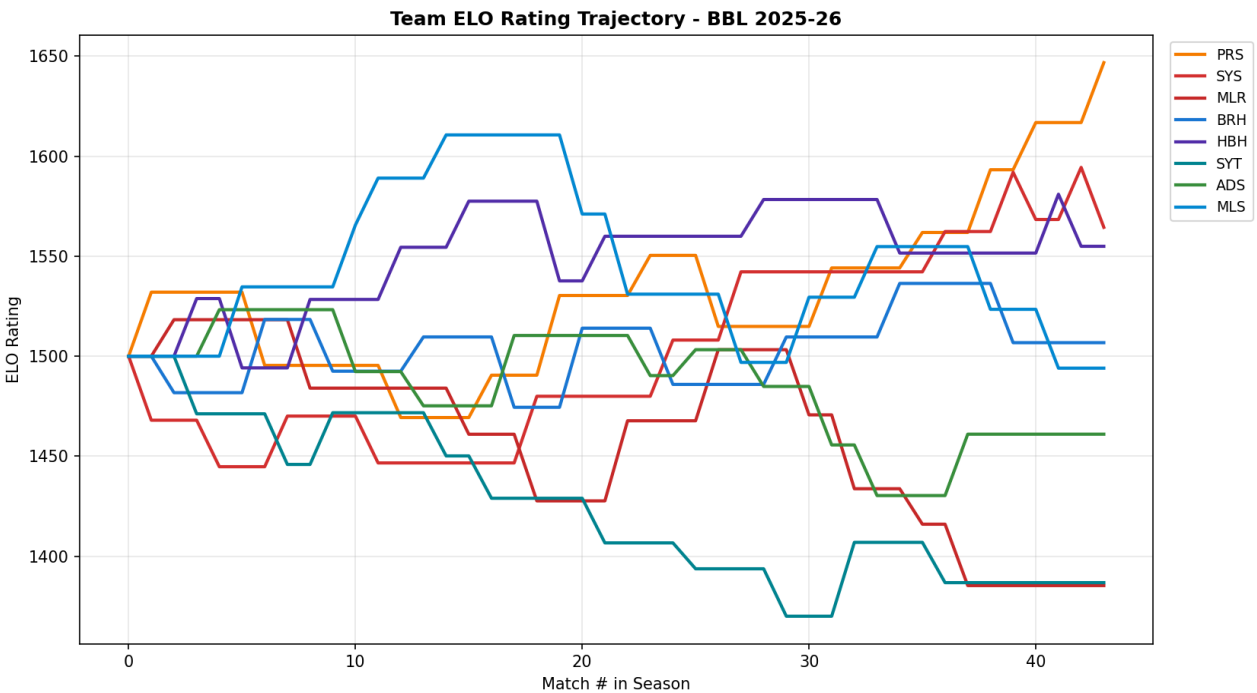


Figure 5: ELO rating trajectory for all eight BBL teams. PRS's championship run is visible as the steady climb at the top; SYT and MLR decline throughout the season at the bottom.

6.1 Final BBL Team Ratings

#	Team	ELO	M	W	Win%	BatRR	BowIRR	Form5
1	PRS	1647	12	9	75.0%	9.38	8.32	100%
2	SYS	1564	12	7	58.3%	8.41	7.97	60%
3	HBH	1555	11	7	63.6%	9.07	8.92	60%
4	BRH	1507	10	5	50.0%	8.91	9.33	60%
5	MLS	1494	11	6	54.5%	8.58	7.82	40%
6	ADS	1461	10	4	40.0%	7.77	8.04	40%
7	SYT	1387	10	2	20.0%	8.62	9.65	20%
8	MLR	1385	10	3	30.0%	8.11	9.22	20%

ELO ratings on January 25, 2026 (after the Final). PRS's 100% last-5 form reflects their strong finish to the season including all three playoff wins.

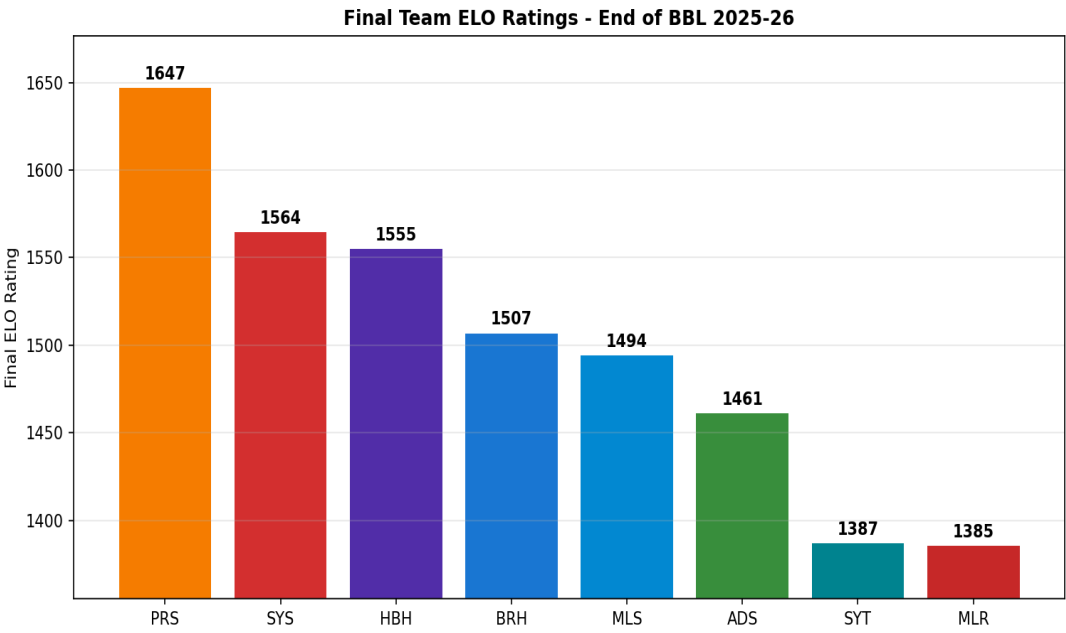


Figure 6: Final ELO ratings at the end of BBL 2025-26.

07 - AUDIT TRAIL

BBL Prediction Log (Winner)

Every prediction made by the BBL-tuned Optimized-Weighted Ensemble is listed below for auditability. Probability is for team A (the team listed first in the schedule). Bold red rows are incorrect predictions.

#	Matchup	P(A)	Winner	Result
1	PRS vs SYS	50%	PRS	OK
2	MLR vs BRH	50%	MLR	OK
3	HBH vs SYT	50%	HBH	OK
4	SYS vs ADS	41%	ADS	OK
5	MLS vs HBH	39%	MLS	X
6	BRH vs PRS	29%	BRH	X
7	SYT vs SYS	53%	SYS	X
8	MLR vs HBH	62%	HBH	X
9	SYT vs BRH	36%	SYT	X
10	ADS vs MLS	45%	MLS	OK
11	SYS vs MLS	32%	MLS	OK
12	PRS vs HBH	45%	HBH	OK
13	BRH vs ADS	47%	BRH	X
14	MLS vs SYT	71%	MLS	OK
15	HBH vs MLR	64%	HBH	OK
16	SYT vs PRS	46%	PRS	OK
17	ADS vs BRH	43%	ADS	X
18	MLR vs SYS	50%	SYS	OK
19	HBH vs PRS	62%	PRS	X
20	BRH vs MLS	27%	BRH	X
21	SYT vs HBH	32%	HBH	OK
22	MLS vs MLR	72%	MLR	X
23	PRS vs ADS	54%	PRS	OK
24	SYS vs BRH	47%	SYS	X
25	ADS vs SYT	62%	ADS	OK
26	PRS vs MLR	62%	MLR	X

#	Matchup	P(A)	Winner	Result
27	MLS vs SYS	57%	SYS	X
28	HBH vs ADS	59%	HBH	OK
29	BRH vs SYT	58%	BRH	OK
30	MLR vs MLS	49%	MLS	OK
31	ADS vs PRS	41%	PRS	OK
32	SYT vs MLR	37%	SYT	X
33	MLS vs ADS	62%	MLS	OK
34	HBH vs BRH	60%	BRH	X
35	MLR vs PRS	36%	PRS	OK
36	SYS vs SYT	72%	SYS	OK
37	ADS vs MLR	50%	ADS	OK
38	PRS vs MLS	54%	PRS	OK
39	BRH vs SYS	42%	SYS	OK
40	PRS vs SYS	52%	PRS	OK
41	HBH vs MLS	51%	HBH	OK
42	SYS vs HBH	51%	SYS	OK
43	PRS vs SYS	55%	PRS	OK

08 - SYNTHESIS

Three-League Conclusions

8.1 The Methodology Is Validated

The Optimized-Weighted Ensemble architecture has now won three consecutive league backtests: IPL 2026 (63.0%), PSL 2026 (67.4%), and BBL 2025-26 (65.1%). Across 159 walk-forward predictions on three continents, the same six-signal weighted blend has outperformed every alternative - including sophisticated ML ensembles. This is no longer an IPL-specific finding; it is a general result about T20 match prediction.

8.2 What Travels vs What Doesn't

The clearest dichotomy from three leagues of evidence: **architecture travels, parameters don't**. The six-signal blend, the ELO rating system with margin-of-victory, the warm-up and retrain cadence - all these design choices are universal. But the optimal weights, the warm-up window size, and even which signals matter most are all league-specific. A production deployment must be willing to re-tune the weights per league, ideally using one full season of historical data.

8.3 The ML Avoidance Rule

Tree-based ML models (Random Forest, Gradient Boosting, Stacked Ensemble) have now failed on three independent leagues. The pattern is structural: small training samples, high match variance, and a low signal-to-noise ratio. These models will overfit, become overconfident on wrong predictions, and underperform the naive baseline. The rule for production: **do not use tree-based ML for seasonal-scale T20 prediction**. They become viable only when hundreds of historical matches are pooled across multiple seasons - which is rarely available.

8.4 League-Specific Tuning Insights

The three leagues reveal a clear tuning pattern that future deployments can use as a starting point. **Higher-scoring leagues with longer seasons** (like IPL) should give less weight to ELO and more to momentum and form - the longer season lets multiple signals stabilise. **Lower-scoring leagues with shorter seasons** (like BBL) should give heavy weight to ELO and little to nothing to momentum - team strength differentials dominate match outcomes, and short schedules prevent momentum from developing.

8.5 Production Deployment Recipe

For a new T20 league, the proven recipe is:

1. Use the six-signal Optimized-Weighted architecture (do NOT substitute ML)
2. Collect at least one full season of historical match data
3. Initialise all teams at ELO 1500 with K=32 and margin-of-victory multiplier
4. Run walk-forward backtest with default IPL weights as starting point
5. Grid-search 500-1000 weight combinations on the backtest horizon
6. Lock the optimal weights for the upcoming season
7. Use 5-match warm-up before generating live predictions
8. After each match, update team states and ELO with the result
9. At season's end, re-tune weights with the new data added

8.6 Limitations & Open Questions

Three honest caveats remain after three leagues. **Sample size:** 159 matches across three leagues is meaningful but still small; the 95% confidence interval on a 65% accuracy is roughly +/- 7 percentage points. **Variance:** T20 cricket remains inherently unpredictable - even the best system will be wrong 35% of the time. **Missing context:** the system still uses only team-level match results - no player availability, venue, toss, or weather data. Adding these on match day would likely add 5-10 percentage points of accuracy on top of the current 65%.

Deliverables accompanying this report: `bbl_predictions.json` (full per-match predictions for all 13 systems on BBL), `bbl_chart_*.png` (4 BBL charts), `chart_three_league_comparison.png` (IPL vs PSL vs BBL), `chart_weight_transfer.png` (cross-league weight penalty), and `bbl_predict.py` (the recoverable analysis script).